Zhidan Liu

Associate Professor
College of Computer Science and Software Engineering
Shenzhen University, Shenzhen, Guangdong, China

Mobile: +86-(755)-86934659
Email: liuzhidan@szu.edu.cn
Homepage: https://liuzhidan.github.io/

Employment

- 2022.07 - Now Associate Professor (tenured), **Shenzhen University**, China
- 2017.07 - 2022.06 Assistant Professor, **Shenzhen University**, China
- 2015.01 - 2017.06 Research Fellow, **Nanyang Technological University**, Singapore (work with *Prof.* **Mo Li**)

Education

- 2009.09 - 2014.09 Ph.D. in Computer Science, **Zhejiang University**, Hangzhou, China, Advisor: *Prof.* **Dongming Lu**
- 2005.09 - 2009.07 B.E. in Computer Science, **Northeastern University**, Shenyang, China

Research Interests

Internet of Things, Urban Computing, Mobile and Ubiquitous Computing, Crowdsourcing, and Big Data Analytics

Honors and Awards

- 2020.12 IEEE ICPADS'20 Best Paper Award
- 2018.06 Shenzhen Nanshan District Pilot Talent Award (深圳市南山区领航人才)
- 2018.01 Shenzhen Peacock Plan Talent Award (深圳市海外高层次人才)

Publications

(† indicates the students I advised or co-advised, * indicates the corresponding author)

Refereed Journal Publications

1. [**TBD**] Biyao Gong†, Tianzhang Xing*, **Zhidan Liu***, Wei Xi, Xiaojiang Chen. "Adaptive Client Clustering for Efficient Federated Learning over Non-IID and Imbalanced Data", *IEEE Transactions on Big Data*, accepted to appear. (IF: 3.344, 中科院大类二区, JCR Q1)

2. [**MONE**] Biyao Gong†, Tianzhang Xing*, **Zhidan Liu***, Junfeng Wang, Xiuya Liu. "Adaptive Clustered Federated Learning for Heterogeneous Data in Edge Computing", *Mobile Networks and Applications*, DOI: https://doi.org/10.1007/s11036-022-01978-8, 2022. (IF: 3.426, 中科院大类三区, JCR Q2)

3. [**CJC**] **Zhidan Liu**, Weixin Lin†, Kaishun Wu, "A Spatial Structure Matching Algorithm for Large Spatial-Textual Datasets / 一种面向大规模空间文本数据的空间结构匹配算法", *Chinese Journal of Computers* (计算机学报), Vol. 45, Issue 6, Pages 1261-1275, June 2022. (CCF Rank-A)

4. [**TIST**] Liang Wang, Zhiwen Yu, Bin Guo, Dingqi Yang, Lianbo Ma, **Zhidan Liu**, Fei Xiong. "Data-Driven Targeted Advertising Recommendation System for Outdoor Billboard", *ACM Transactions on Intelligent Systems and Technology*, Vol. 13, No. 2, Article 29, Pages 1-23, January 2022. (IF: 4.654, 中科院大类三区, JCR Q1)

5. [**IOTJ**] **Zhidan Liu**, Zengyang Gong†, Jiangzhou Li†, Kaishun Wu, "mT-Share: A Mobility-Aware Dynamic Taxi Ridesharing System", *IEEE Internet of Things Journal*, Vol. 9, Issue 1, Pages 182-198, January 2022. (IF: 9.936, 中科院大类一区, JCR Q1)

6. [**TITS**] **Zhidan Liu**, Jiangzhou Li†, Kaishun Wu. "Context-Aware Taxi Dispatching at City-Scale Using Deep Reinforcement Learning", *IEEE Transactions on Intelligent Transportation Systems*, Vol. 23, Issue 3, Pages 1996-2009, March 2022. (IF: 6.319, CCF Rank-B, 中科院小类一区, JCR Q1)

7. [**IOTJ**] Shiwei Song†, **Zhidan Liu**, Zhenjiang Li, Tianzhang Xing, Dingyi Fang. "Coverage-Oriented Task Assignment for Mobile Crowdsensing", *IEEE Internet of Things Journal*, Vol. 7, Issue 8, Pages 7407-7418, April 2020. (IF: 9.936, 中科院大类一区, JCR Q1)

8. [**IOTJ**] **Zhidan Liu**, Zhenjiang Li, Kaishun Wu. "UniTask: A Unified Task Assignment Design for Mobile Crowdsourcing based Urban Sensing", *IEEE Internet of Things Journal*, Vol.6 Issue 4, Pages 6629-6641, August 2019. (IF: 9.936, 中科院大类一区, JCR Q1)

9. [**TMC**] **Zhidan Liu**, Pengfei Zhou, Zhenjiang Li, Mo Li. "Think Like A Graph: Real-Time Traffic Estimation at City-Scale", *IEEE Transactions on Mobile Computing*, Vol. 18, Issue 10, Pages 2446-2459, October 2019. (IF: 5.112, CCF Rank-A, 中科院大类二区, JCR Q1)

10. [**IEEENet**] **Zhidan Liu**, Zhenjiang Li, Kaishun Wu, Mo Li. "Urban Traffic Prediction from Mobility Data Using Deep Learning", *IEEE Network Magazine*, Vol. 32, Issue 4, Pages 40-46, August 2018. (IF: 8.808, 中科院大类一区, JCR Q1)

11. [**TITS**] **Zhidan Liu**, Shiqi Jiang, Pengfei Zhou, and Mo Li. "A Participatory Urban Traffic Monitoring System: The Power of Bus Riders", *IEEE Transactions on Intelligent Transportation Systems*, Vol. 18, Issue 10, Pages 2851 - 2864, October 2017. (IF: 4.051, CCF Rank-B, 中科院小类一区, JCR Q1)

12. [**TON**] **Zhidan Liu**, Zhenjiang Li, Mo Li, Wei Xing, and Dongming Lu. "Path Reconstruction in Dynamic Wireless Sensor Networks Using Compressive Sensing", *IEEE/ACM Transactions on Networking*, Vol. 24, Issue 4, Pages 1948 - 1960, August 2016. (IF: 3.376, CCF Rank-A, 中科院大类二区, JCR Q1)

13. [**TITS**] **Zhidan Liu**, Zhenjiang Li, Mo Li, Wei Xing, and Dongming Lu. "Mining Road Network Correlation for Traffic Estimation via Compressive Sensing", *IEEE Transactions on Intelligent Transportation Systems*, Vol. 17, Issue 7, Pages 1880 - 1893, July 2016. (IF: 3.724, CCF Rank-B, 中科院小类一区, JCR Q1)

14. [**IJDSN**] Bo Zeng, Yabo Dong, **Zhidan Liu**. "A Negotiation-Based TDMA Scheduling with Consecutive Slots Assignment for Wireless Sensor Networks", *International Journal of Distributed Sensor Networks*, 2014. (IF: 1.239, 中科院大类四区, JCR Q4)

15. [**IJDSN**] **Zhidan Liu**, Wei Xing, Yongchao Wang, Dongming Lu. "Hierarchical Spatial Clustering in Multihop Wireless Sensor Networks", *International Journal of Distributed Sensor Networks*, 2013. (IF: 1.239, 中科院大类四区, JCR Q4)

Refereed Conference Publications

1. [**DASFAA'21**] **Zhidan Liu**, Junhong Zheng†, Zengyang Gong†, Haodi Zhang, Kaishun Wu. "Exploiting Multi-Source Data for Adversarial Driving Style Representation Learning", In *DASFAA*, Taipei, Taiwan, April, 2021. (CCF Rank-B)

2. [**HPCC'20**] Baofu Huang†, **Zhidan Liu***, Kaishun Wu. "Structure Preserved Graph Reordering for Fast Graph Processing Without the Pain", In *IEEE HPCC*, Fiji, December, 2020. (Acceptance Rate: 58/379 = 15%, CCF Rank-C)

3. [**ICPADS'20**] Baofu Huang†, **Zhidan Liu***, Kaishun Wu. "Accelerating PageRank in Shared-Memory for Efficient Social Network Graph Analytics", In *IEEE ICPADS*, HongKong, China, December, 2020. (**Best Paper Award**, CCF Rank-C)

4. [**ICCCN'20**] Zengyang Gong†, Bo Du, **Zhidan Liu**, Wei Zeng, Pascal Perez, Kaishun Wu. "SD-seq2seq : A Deep Learning Model for Bus Bunching Prediction Based on Smart Card Data", In *IEEE ICCCN*, Honolulu, Hawaii, USA, August 2020. (CCF Rank-C)

5. [**ICDE'20**] **Zhidan Liu**, Zengyang Gong†, Jiangzhou Li†, Kaishun Wu. "Mobility-Aware Dynamic Taxi Ridesharing", In *IEEE ICDE*, Dallas, Texas, USA, April 2020. (CCF Rank-A)

6. [**MSN'19**] **Zhidan Liu**, Mingliang Huang†, Zhi Ye†, Kaishun Wu. "DeepRTP: A Deep Spatio-Temporal Residual Network for Regional Traffic Prediction", In *IEEE MSN*, Hong Kong, China, December 2019. (Invited paper, CCF Rank-C)

7. [**MobiSys'19**] Yang Liu, Zhenjiang Li, **Zhidan Liu**, Kaishun Wu. "Real-time Arm Skeleton Tracking and Gesture Inference Tolerant to Missing Wearable Sensors", In *ACM MobiSys*, Seoul, Republic of Korea, June 2019. (Acceptance Rate: 40/174 = 22%, CCF Rank-B, CSRanking Conference)

8. [**IPSN'18**] Chu Cao, **Zhidan Liu**, Mo Li, Qin Zheng, Wenqiang Wang. "Walkway Discovery from Large Scale Crowdsensing", In *ACM/IEEE IPSN*, Porto, Portugal, April 2018. (Acceptance Rate: 22/82 = 26%, CCF Rank-B)

9. [**MobiHoc'14**] **Zhidan Liu**, Zhenjiang Li, Mo Li, Wei Xing, Dongming Lu. "Path Reconstruction in Dynamic Wireless Sensor Networks using Compressive Sensing", In *ACM MobiHoc*, Philadelphia, USA, August 2014. (Acceptance Rate: 40/211 = 18%, CCF Rank-B)

10. [**ICCC'13**] **Zhidan Liu**, Wei Xing, Yongchao Wang, Dongming Lu. "Hierarchical Spatial Clustering in Multi-hop Wireless Sensor Networks", In *IEEE ICCC*, Xi'an, China, August 2013.

11. [**AINA'13**] **Zhidan Liu**, Wei Xing, Bo Zeng, Yongchao Wang, Dongming Lu. "Distributed Spatial Correlation-based Clustering for Approximate Data Collection in WSNs", In *IEEE AINA*, Barcelona, Spain, March 2013.

12. **[ICACT'13] Zhidan Liu**, Wei Xing, Yongchao Wang, Dongming Lu. "An Energy-Efficient Data Collection Scheme for Wireless Sensor Networks", In *IEEE ICACT*, PyeongChang, Korea, January 2013.

13. **[CyberC'12]** Bo Zeng, Yabo Dong, **Zhidan Liu**, Dongming Lu. "A Workload-Aware Link Scheduling for Heterogeneous Wireless Sensor Networks", In *IEEE CyberC*, Sanya, China, October 2012.

14. **[WiCom'12] Zhidan Liu**, Wei Xing, Yongchao Wang, Dongming Lu. "Adaptive Data Storage for Querying in Wireless Sensor Networks", In *IEEE WiCom*, Shanghai, China, September 2012.

Books, Poster, and Demo

1. **[Book'21] Zhidan Liu**, Kaishun Wu. "Mobility Data-Driven Urban Traffic Monitoring", *Springer*, Singapore, May, 2021.

2. **[MobiSys'19]** Yang Liu, Chengdong Lin, Zhenjiang Li, **Zhidan Liu**, Kaishun Wu. "Poster: WhenWearable Sensing Meets Arm Tracking", In *ACM MobiSys*, Seoul, Republic of Korea, June 2019.

3. **[IPSN'18]** Chu Cao, **Zhidan Liu**, Mo Li, Qin Zheng, Wenqiang Wang. "Demo Abstract: Walkway Discovery from Large Scale Crowdsensing", In *ACM/IEEE IPSN*, Porto, Portugal, April 2018.

Patents

1. **刘志丹**, 李江舟, 伍楷舜. 一种基于深度强化学习的出租车调度方法及系统 (ZL202010521882.1), 2021.07, 中国发明专利

2. **刘志丹**, 周鹏飞,李镇江, 李默, 伍楷舜. 城市路网交通估计方法及系统 (ZL201811140012.9), 2021.09, 中国发明专利

Selected Projects/Grants

- **[PI] Research on the Key Technologies of Mobility Context Aware Personalized Ridesharing** / 基于移动情境感知的个性化共享出行关键技术研究. 国家自然科学基金 (NSFC) - 面上项目, Grant No.62172284, 2022.01 - 2025.12, 590K CNY. (Ongoing)

- **[PI] Research on the Key Technologies of Shared Memory based Social Network Graph Computing** / 面向共享内存的社交网络图谱计算关键技术研究. 广东省自然科学基金 - 面上项目, Grant No.2022A1515010155, 2022.01 - 2024.12, 100K CNY. (Ongoing)

- **[PI] Joint Research of Crowdsensing and Data Analytics for Smart City** / 面向智慧城市的群智感知与数据分析技术研究. 广东省自然科学基金 - 面上项目, Grant No.2020A1515011502, 2019.10 - 2022.09, 100K CNY. (Ongoing)

- **[PI] Research on Exploiting Big Mobility Trajectory Data for Smart Sensing of Urban Traffic** / 基于移动轨迹大数据的城市交通智能感知研究. 国家自然科学基金 (NSFC) - 青年项目, Grant No.61802261, 2019.01 - 2021.12, 250K CNY. (Completed)

- **[PI] Research on Big Mobility Trajectory Data based Intelligent Transport Systems** / 基于移动大数据的智能交通研究. 深圳市高端人才(孔雀计划)科研启动项目, Grant No.827/000344, 2018.01 - 2021.12, 2700K CNY. (Completed)

- **[PI] Research on Exploiting Big Trajectory Data for Intelligent Transport Sensing and Computing** / 基于轨迹大数据的智能交通感知与计算的研究. 腾讯"犀牛鸟"-深圳大学青年教师科研基金项目, Grant No.201802, 2019.01 - 2020.12, 50K CNY. (Completed)

- **[Co-PI] Multi-dimensional and Deep Perception Information based Social Computing** / 基于多维度深层次感知信息的社群情境计算. 国家自然科学基金 (NSFC) - 联合重点基金, Grant No.U2001207, 2021.01 - 2024.12, 2550K CNY. (Ongoing)

Academic Community Service

Membership:

- The Institute of Electrical and Electronics Engineers (IEEE), since 2015
- The Association for Computing Machinery (ACM), since 2018
- China Computer Federation (CCF), since 2020

Conference Organization Chair:

- Special session Chair: IEEE CSCWD 2022
- Local Chair: EAI ICECI 2021
- Conference Finance Chair: IEEE ICPADS 2017
- Conference Finance Chair: IEEE ICPADS 2018

Technical Program Committee Member:

- 2022: ACM KDD, ACM WSDM, IEEE ICPADS
- 2021: IEEE ICDCS, IEEE MASS
- 2020: IEEE ICDCS, IEEE HiPC
- 2019: IEEE MASS, IEEE MSN
- 2018: IEEE ICPADS, IEEE ICCCN
- 2017: IEEE ICPADS

Transaction and Journal Reviewer (selected):

- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- IEEE Transactions on Mobile Computing (TMC)
- IEEE Transactions on Intelligent Transportation Systems (TITS)
- IEEE/ACM Transactions on Networking (TON)
- ACM Transactions on Spatial Algorithms and Systems (TSAS)
- IEEE Transactions on Wireless Communications (TWC)
- IEEE Transactions on Parallel and Distributed Systems (TPDS)
- ACM Transactions on Sensor Networks (TOSN)
- IEEE Transactions on Communications (TOC)
- IEEE Internet of Things Journal (JIOT)
- IEEE Communications Magazine
- IEEE Computational Intelligence Magazine
- Elsevier Pervasive and Mobile Computing (PMC)
- Ad Hoc & Sensor Wireless Networks (AHSWN)
- Elsevier Computer Networks

Teaching

Teaching Courses:

- Introduction to Wireless Sensor Networks 无线传感器网络 Since 2017 (Semester-1)
- Internet Programming (in JAVA) 互联网编程 Since 2017 (Semester-2)

Teaching Achievements:

- 2022.06 指导的本科生刘键聪获得**"深圳大学2022届本科优秀毕业生"**称号
- 2022.06 指导本科生刘键聪获得**"深圳大学2022届百篇优秀本科毕业论文(设计)"**
- 2021.12 被评为深圳大学**"新锐研究生导师"**
- 2021.09 被评为2021年广东省大学生计算机设计大赛"优秀指导教师"
- 2021.08 指导本科生张柏林, 许伟鑫, 郑元初获得2021年(第14届)**中国大学生计算机设计大赛全国二等奖**
- 2021.07 指导本科生刘键聪等人成功申请到**广东省省级大学生创新创业训练计划项目一**项(项目名"基于深度学习的GPS定位辅助纠正系统"), 经费10000元 (项目结题: 良好, 2022.06.30)
- 2021.05 获得深圳大学第十届青年教师讲课竞赛学院"二等奖"
- 2021.04 指导硕士研究生周荧倩等人成功申请到**深圳大学研究生创新发展基金项目(自然科学)一**项(项目名"面向时空大数据的查询与分析技术研究"), 经费8000元 (项目结题: 优秀, 2022.07.15)
- 2021.04 获得深圳大学2019-2020学年教学单项奖 - **"优秀本科教师奖"**
- 2020.12 获得2020年深圳大学腾讯益友奖**"优秀班主任"**
- 2020.12 获得2020年(第13届)中国大学生计算机设计大赛"星级优秀指导教师"称号
- 2020.08 指导本科生陈林桦, 李宇良, 刘键聪获得2020年(第13届)**中国大学生计算机设计大赛全国一等奖**
- 2020.06 指导的硕士研究生龚增阳获得深圳大学**"2020届优秀毕业研究生"**
- 2019.06 获得深圳大学第九届青年教师讲课竞赛学院"二等奖"
- 2019.04 指导本科生黄柱邦, 钟金成, 杨虎林获得2019年美国大学生数学建模竞赛(MCM/ICM)二等奖 (H奖)

Research Group

Current Students:

- **Meiquan Lai** (赖美全), (Master student, 2020.09-)
  B.S. in Mathematics, Shenzhen University, Shenzhen, China, 2020
- **Yingqian Zhou** (周荧倩), (Master student, 2020.09-)
  B.S. in Computer Science and Technology, Guangdong University of Technology, Guangzhou, China, 2020
- **Hongquan Zhang** (张鸿权), (Master student, 2020.09-)
  B.S. in Internet of Things Engineering, Hunan Agricultural University, Changsha, China, 2020
- **Jiayue Ruan** (阮家越), (Master student, 2020.09-)
  B.S. in Computer Science and Technology, North China University of Technology, Beijing, China, 2020
- **Yaqi Huang** (黄雅琪), (Master student, 2021.09-)
  B.S. in Internet of Things Engineering, Hunan Agricultural University, Changsha, China, 2021
- **Guofeng Ouyang** (欧阳国峰), (Master student, 2021.09-)
  B.S. in Electronical Information Science and Technology, Zhaoqing University, Zhaoqing, China, 2021
- **Jinye Lin** (林锦叶), (Master student, 2021.09-)
  B.E. in Internet of Things Engineering, Shenzhen University, Shenzhen, China, 2021
- **Bolin Zhang** (张柏林), (Undergraduate student, 2019-)
- **Wenxuan Xia** (夏文轩), (Undergraduate student, 2020-)
- **Xiaowen Xu** (徐小雯), (Undergraduate student, 2021-)
- **Haotian Zhong** (钟昊天), (Undergraduate student, 2022-)

Alumni:

1. **Zengyang Gong** (龚增阳), (Master 2017.09 - 2020.06)
   - Awards: 深圳大学2020届"优秀毕业研究生"
   - Publications: IEEE ICDE'20, IEEE ICCCN'20, IEEE IOTJ'22
   - Employment: Ph.D student in The Hong Kong university of Science and Technology (HKUST)
   - B.S. in Computer Science and Technology, Wuhan University of Science and Technology, Wuhan, China, 2017
2. **Mingliang Huang** (黄明亮), (Master 2017.09 - 2020.06)
   - Publications: IEEE MSN'19
   - Employment: Sangfor Technologies (深信服)
   - B.S. in Computer Science and Technology, Lanzhou University, Lanzhou, China, 2017
3. **Jiangzhou Li** (李江舟), (Master 2018.09 - 2021.06)
   - Publications: IEEE Trans. ITS'20
   - Employment: Sangfor Technologies (深信服)
   - B.S. in Software Engineering, Qingdao University, Qingdao, China, 2018
4. **Baofu Huang** (黄保福), (Master 2018.09 - 2021.06)
   - Awards: "Best Paper Award" of IEEE ICPADS'20
   - Publications: IEEE HPCC'20, IEEE ICPADS'20
   - Employment: Tencent (腾讯)
   - B.E. in Computer Science, Guangdong University Of Finances and Economics, Guangzhou, China, 2018
5. **Weixin Lin** (林维鑫), (Master 2019.09 - 2022.06)
   - Publications: CJC'22
   - Employment: JinTeng Technologies (金腾科技)
   - B.E. in Computer Science and Technology, Shenzhen University, Shenzhen, China, 2019
6. **Junhong Zheng** (郑俊鸿), (Master 2019.09 - 2022.06)
   - Publications: DASFAA'21
   - Employment: Tencent (腾讯)
   - B.S. in Software Engineering, Zhuhai College of Science and Technology, Zhuhai, China, 2019

7. **Jiancong Liu** (刘键聪), (B.S. 2018.09 - 2022.06)
   - Awards: 深圳大学2022届“本科优秀毕业生”, 深圳大学2022届百篇优秀本科毕业论文
   - Employment: Master student in Harbin Institute of Technology (Shenzhen)
   - B.E. in Computer Science and Technology, Shenzhen University, Shenzhen, China, 2022

Updated in July, 2022